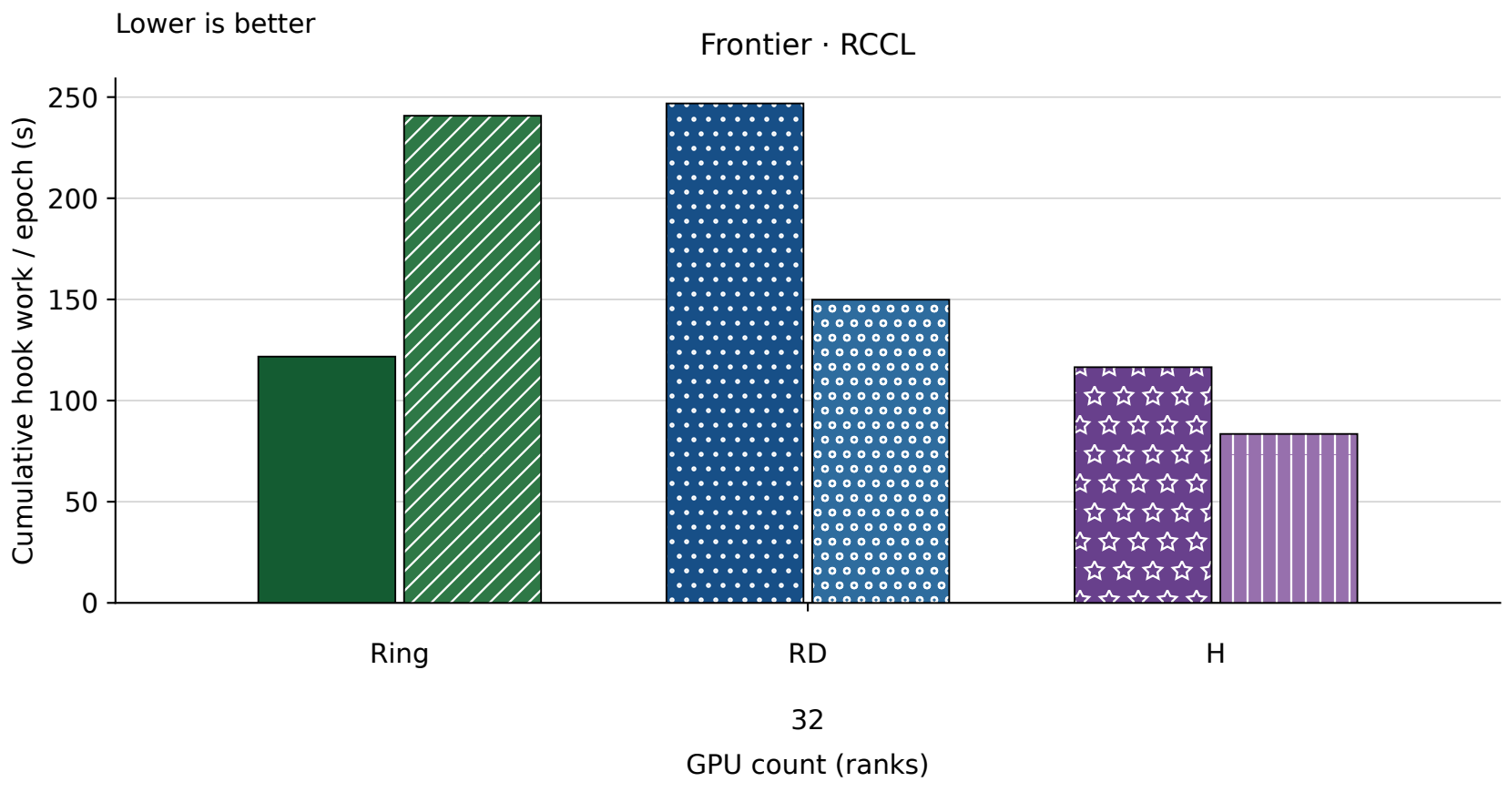

WideResNet 32×32 · strong scaling · effective global batch 128
Lower is better
Frontier · RCCL
Cumulative hook work / epoch (s)
250
200
150
100
50
0
Ring
RD
H
32
GPU count (ranks)
Ring
Uncompressed
ZFP naive · rate16
Recursive doubling
Uncompressed
ZFP naive · rate16
Hierarchical
ZFP online · rate16
ZFP online · rate8
Epoch 1 excluded · missing metrics are not zero · r = ZFP rate